

Regression-based Finish Time Prediction of 100km Trail Running using Structured Athlete Performance

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Abstract—Finish times prediction in trail running offers valuable support to athletes, coaches, and event organizers in planning and performance assessment. This study presents a benchmarking analysis using a newly compiled exploratory dataset comprising 100 trail runners. The benchmark dataset consists of 18 structured features, including demographic attributes (age and gender), performance indices (ITRA and UTMB scores), race history (number of 50km and 100km events), and recent performance times. Time-based features were normalized, and missing values were replaced using sentinel values and binary indicators. Furthermore, the study introduces the integration of recent race finish times alongside elevation gain to enhance model accuracy. Thirteen regression-based models were evaluated to explore the predictive power of these features. Among them, Support Vector Regression (SVR) demonstrated the better results compared to other regression models on the actual race dataset, with an R^2 score close to zero (-0.018), 256.25 minutes on Mean Absolute Error (MAE), and 294.84 minutes on Root Mean Squared Error (RMSE). These results provide promising indicators in race time prediction using limited data and contribute to future research in endurance sports analytics.

Keywords— trail running, finish time, prediction, regression models

I. INTRODUCTION

Trail running is a growing endurance sport that presents unique challenges due to its varied terrain, elevation gain, and unpredictable environmental conditions. Predicting finish time in trail running events can be beneficial for athletes, coaches, and event organizers, especially in planning strategies, pacing, and logistics. However, accurate prediction remains difficult due to the complexity of the sport and the limited structured dataset.

One of the main challenges for predicting the finish time is the difficulty of collecting detailed individual-level data, such as training logs, physiological metrics, or wearable device outputs. These types of data are often private, inconsistent, or require direct cooperation from athletes, which limits scalability and reproducibility in research. To overcome this limitation, our study focuses on using publicly available data from platforms such as UTMB, ITRA, and the TMBT (The Most Beautiful Thing) website, which provide public structured race results and athlete indices. However, the data collection is still manual and often time-consuming.

This study aims to develop a benchmark dataset that incorporates structured athlete profiles and performance history, using features that are both meaningful and accessible. The dataset includes demographic information (age, gender), performance indices (ITRA, UTMB, UTMB100), race history (number of 50km and 100km

events), and recent race performance metrics. Time-based features are normalized to account for variability, and elevation data is included to reflect race difficulty. The target variable is the Race 100KM Time (in minutes).

The objectives of this study are (a) to prepare a benchmark dataset using structured and publicly available features relevant to trail running performance, and (b) to explore the predictive power of these features using regression-based models and evaluate model performance across multiple metrics.

By focusing on accessible data and practical features, this study provides a scalable approach to race time prediction and contributes to the development of data-driven tools in endurance sports analytics. Additionally, benchmarking the performance of regression models on this newly constructed dataset will provide initial insights into the feasibility of race time prediction with limited but meaningful data.

II. LITERATURE REVIEW

A. Related Works

Table I shows the summary of the studies in predicting the race finish time according to feature demographics, modelling approach, and performance or result from the metric used. Among the studies, the regression model, specifically the XG Boost, and a study comparing Artificial Neural Network (ANN) and k-Nearest Neighbour (KNN). This provides an opportunity to explore other machine learning algorithms for comparison. In a feature used and dataset perspective, all of the studies use very individual-specific information, e.g., collected from runners, races, and may use specific tools or equipment (lactate value, etc.).

Most studies use R-squared (R^2), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE) as evaluation metrics in the prediction. MAE is the measure of the average absolute difference between predicted and actual finish time, thus can be used to interpret the prediction error in minutes or hours. RMSE shows the outliers and penalizes the larger errors more than MAE. This is important to assess how the model handles extreme finish times given certain circumstances. The R^2 indicates the proportion of variance in the target variable (race time) and is often a standard metric for regression models. Additionally, other metrics were also used, namely, SHapley Additive exPlanations (SHAP), p-values, regression coefficients, and feature importance scores. Regression Coefficients are values representing the relationship between input features and the predicted race time.

TABLE I. SUMMARY OF RELATED WORKS IN PREDICTING RACE FINISH TIME

Study	Features Used (Dataset, Sample Size)	Modeling Approach	Performance / Results (Metrics)
Western States Endurance Run 161km (Wegelin and Hoffman, 2011) [1]	Western States Endurance Run (1986-2007), 576 finishes; Age, sex, race history (1st time starter status, prior finish rate, prior relative performance)	Logistic regression (finish odds); linear regression (finish time)	Focused on regression coefficients, odds ratios, and significance levels, No R^2 /MAE/RMSE reported.
Suter et al. 2020 (UTMB 2008–2019) [2]	UTMB official result (2008–2019); 15902 finishers; Age, sex, race time, section split times (checkpoint)	Statistical analyses (Descriptive stats, ANOVA, correlations)	Even pacing = faster times; correlation/effect sizes/ $p < 0.05$; no direct RMSE.
10-Day Ultramarathon (Knechtle et al., 2024) [3]	10-day ultra races (1991–2022); 452 runners, 6 countries; Age group, gender, country, surface	XG Boost regression	Outcome (speed km/h), Best model (200 estimators, learning rate=0.5), $R^2 = 0.65$, MAE = 0.37 km/h.
72-Hour Ultramarathon (Knechtle et al., 2024) [4]	Global 72-h ultra races; $N \approx 1,500$; Age, gender, event country, elevation	Gradient boosted trees	$R^2 \approx 0.61$, RMSE ≈ 0.48 km/h.
48-Hour Ultramarathon (Knechtle et al., 2025, <i>Sci Reports</i>) [5]	Global 48-h races; 16,233 race records (60 countries held in 36 countries); Age, gender, country, course features	ML feature-importance analysis (XGBoost regression model to predict running speed)	1500 estimators, learning rate=0.5, $R^2 \approx 0.49$, MAE ≈ 0.77 km/h; course characteristics strongest (terrain/surface) predictors.
Scheer et al. (Short Trail Study) (2019) [6]	Trail race (~31 km, 1800m elevation gain); 9 trained male runners; Lactate threshold, VO_{2max} , demographics	Linear regression to identify the strongest predictors	Strongest predictor (speed), VO_{2max} is not a significant predictor, the regression model for individual anaerobic threshold (IAT), $R^2 = 0.86$, in trail race performance time.
Vickers et al. (Recreational Running) (2016) [7]	Recreational runners (5k–marathon); 2303 recreational endurance runners; features (Self-reported training (weekly mileage, long run distance, training	Regression models predicting race times (5km, 10km, half-marathon, marathon)	Best predictors: previous race times, training variables improved the prediction. No R^2 /MAE/RMSE reported.

	frequency, speed work), demographics (age, sex, BMI), prior times.)		
Lerebourg et al. (Marathon Prediction) (2023) [8]	Marathon races, French ranking for 10km road and marathon in 2019, aged 21; 820 runners; features (10km time, BMI, age, gender, marathon time)	ANN vs KNN as a regression model to predict marathon finish time.	ANN MAE=5.6% (4 min 48s) vs KNN MAE=2.4% (11 min 16s); ANN more precise.
Knechtle et al. (Elevation to predict 100km performance) [9]	Data between 1892-2022 (317,312 runners, 103 countries, 100km races); Features (Gender, age group, country of origin of the athlete, country where the race was held, course characteristics)	XGBoost regression was used to assess the predictors of average running speed in (km/h).	$R^2 = 0.51$, MAE (km/h) = 1.22. Elevation is the most important feature.

It is used to understand how many features contribute to the prediction. SHAP can explain the output of certain models by indicating how each feature contributes to a prediction. The feature importance score can be used to rank features according to their contribution to a model's overall accuracy and thus may be used for feature selection.

From the review of the past studies as listed in Table I, we can see that most researchers used methods like linear regression, XGBoost, and other tree-based models to predict finish times or running speed. While these models have demonstrated varying degrees of predictive accuracy, such as R^2 and MAE, they often rely on demographic and environmental features, with limited integration of structured athlete performance data.

B. Regression models for finish time prediction

A summary of the regression models that can be used related to race finish time prediction is listed in Table II, outlining their model and reference, description, and strength.

Based on the past studies listed in Table II, researchers have explored a wide range of regression models to predict race performance, each with its own strengths. Regression models such as Tree-based models like XGBoost, Random Forest, and Gradient Boosting are known for handling non-linear relationships and provide high accuracy, especially on structured data.

Linear regression has been shown to provide the strongest predictors given specific individual-level data. Regularized linear regression models such as Ridge, Lasso, and ElasticNet offer simplicity and interpretability but may struggle with complex patterns.

TABLE II. SUMMARY OF REGRESSION MODELS AND THEIR STRENGTH

Model (Reference)	Description	Strengths
XGBoost Regression Grekousis (2025) [10]	A tree-based ensemble model that builds sequential learners to minimize error.	High accuracy, handles missing data, interpretable via SHAP values.
Linear Regression Heuberger et al. (2018) [11]	Assume input features and race finish time correlate in a linear pattern.	Simple, fast, easy to interpret.
Ridge & Lasso Regression Enwere et al. (2023) [12]	Regularized linear models that reduce overfitting and improve feature selection.	Handles multicollinearity, improves generalization.
Random Forest Regression He et al. (2018) [13]	An ensemble of decision trees that averages predictions.	Captures non-linear relationships, robust to noise and overfitting.
Support Vector Regression (SVR) Rivas-Perea et al. (2013) [14]	Uses kernel functions to model complex relationships with a margin of tolerance.	Effective in high-dimensional spaces, robust to outliers.
Gradient Boosting Regression Bentéjac et al. (2021) [15]	Builds models sequentially to minimize prediction error.	High performance on structured data, customizable.
Neural Network Regression (ANN) Fatima and Johrendt (2025) [16]	Deep learning model with multiple layers capturing complex patterns.	Powerful for large datasets, models non-linear relationships.
K-Nearest Neighbors (KNN) Regression Halder et al. (2024) [17]	Prediction is based on k-nearest neighbour instances.	Non-parametric, simple, good for pattern-based data.
Bayesian Regression Silva et al. (2019) [18]	Probabilistic model incorporating prior beliefs and uncertainty.	Confidence intervals, suitable for small or incomplete datasets.
Decision Tree Costa & Pedreira (2023) [19]	A non-parametric model that splits data into branches based on feature thresholds, forming a tree structure. Each leaf represents a prediction.	Has interpretable results, handle mixed data, robust to outliers.
Bayesian Ridge Silva et al. (2021) [20]	A linear regression model with Bayesian regularization. It applies Gaussian priors to coefficients, allow probabilistic interpretation.	Provides uncertainty estimates, suitable for small datasets
ElasticNet Tay et al. (2021) [21]	Variable selection and regularization are based on L1 (Lasso) and L2 (Ridge) penalties.	Balances feature selection, effective in high-dimensional data
Gradient Boosting Rizkallah (2025) [22]	An ensemble method that builds models sequentially, each correcting the errors of the previous one using gradient descent.	Handles complex non-linear relationships
LightGBM Du et al. (2025) [23]	Based on a gradient boosting framework.	Fast, scalable, handles large datasets, supports categorical features natively.

Neural networks (ANN) and KNN regression are powerful for large or pattern-based datasets but can be

sensitive to noise and require careful tuning, and were specifically tested on very individual-level performance data (Lactate threshold, VO₂max, BMI, etc.). Bayesian models provide uncertainty estimates, making them suitable for small or incomplete datasets.

Among these, Support Vector Regression (SVR) stands out for its ability to model non-linear relationships using kernel functions, making it effective in high-dimensional spaces and robust to outliers. This makes SVR a strong candidate for predicting 100km trail running finish times, especially when using structured athlete performance data that includes UTMB and ITRA indexes, race times, and terrain features, which are the elements that previous studies have not included.

III. METHODOLOGY

A. Data Collection and Feature Selection

This study focuses on predicting finish time for 100km trail running events using structured athlete data and artificial intelligence models. The dataset was constructed using publicly available sources, including UTMB and ITRA databases, race organizer participant lists, and published race results from the race organizer. The target race in which this study was conducted was The Most Beautiful Thing 2025 (TMBT 2025) for the 109km category.

A total of 100 athlete profiles were collected, comprising 88 male and 12 female runners. All selected athletes had a UTMB index of at least 200, ensuring a minimum level of trail running experience and performance consistency. The features selected for this study include demographic information, historical race performance, and race-specific elevation data. The prepared benchmark dataset (100 records) is small, but this study explores the available data for predicting the finish race time using regression models. The details of the 18 features are listed in Table III.

All of the data in this study were obtained from publicly accessible sources and did not contain any personally identifiable information (PII). Athlete names, bib numbers, and other sensitive identifiers were excluded during data extraction. The dataset was compiled only for research purposes and is not shared or distributed beyond the scope of this study. This approach complies with ethical standards for secondary data use, ensuring privacy protection and responsible data handling. No direct contact was made with athletes, and no consent was required due to the public nature and anonymization of the data.

TABLE III. LIST OF FEATURES

Features	Description
Demographic (2)	Age (Mean of category according to UTMB), Gender
Performance index (3)	ITRA Index, UTMB Index, UTMB100 Index (index may change according to the updates after the recent event)
Race history (3)	Number of 50km, Number of 100km, Has recent 100KM
Performance times (4)	Recent 50KM, Recent 100KM, Best 100KM, Longest 100KM
Race-specific (1)	Recent race 100KM Elevation Gain
Normalized features (4)	Recent 50KM Normalized, Recent 100KM Normalized, Best 100KM Normalized, Longest 100KM Normalized
Target variable (1)	Race 100KM Time

B. Preprocessing

To ensure consistency across the dataset, preprocessing on the collected data is as follows:

- All PII's were removed from the dataset
- All time-based features were converted to minutes.
- Missing values were handled using sentinel values (e.g., -1 for no prior race participation and no timing).
- Binary indicators were used for categorical flags such as Has Recent 100KM.
- Normalization was applied to time features to reduce scale variance and improve model learning.

C. Experiment Setup

In the experiment setup, two main experiments were conducted, with Experiment 1 aimed at finding out the best regression model that will be used in Experiment 2. Detail of the experiment is listed in Table IV.

In Experiment 1, regression models were trained using the 80-20 splitting approach to predict the 100KM race finish time dataset. In Experiment 2, the actual race finish time from the target event was collected and used solely for evaluating the prediction accuracy of the models trained in Experiment 1. No retraining was performed, ensuring a fair comparison between predicted and actual performance.

D. Model Selection and Evaluation

Thirteen regression models were tested to evaluate their predictive performance, including Artificial Neural Network (ANN), Bayesian Regression, Decision Tree, ElasticNet, Gradient Boosting, K-Nearest Neighbors (KNN) Regression, LightGBM, Linear Regression, Random Forest Regression, Ridge and Lasso Regression, Support Vector Regression (SVR), and XGBoost Regression.

Furthermore, each model was evaluated using the popular metrics such as R^2 to measure variance in race finish time, MAE to get the average prediction error in minutes, and RMSE, which evaluates the larger errors penalty and highlights model sensitivity.

IV. RESULTS

A. Experiment 1

In this experiment, thirteen regression models were trained and evaluated using an 80-20 split on the benchmark dataset. The goal was to predict the 100KM trail running race time using the selected athlete features. The models were primarily assessed using three key metrics: R^2 , MAE, and RMSE. Table V outlines the findings.

Based on the result, the Linear Regression achieved the highest performance with an R^2 of 1.000. This indicates a perfect fit on the test data. The extremely low MAE (0.027) and RMSE (0.039) suggest minimal prediction error. This result may reflect strong linear relationships in the dataset, especially between normalized time features and the target race time. Lasso Regression also performed exceptionally well ($R^2 = 0.998$), with slightly higher error values.

TABLE IV. EXPERIMENT SETUP

Experiment	Description	Results
1	Training uses 80-20 splitting for each regression model experiment.	Metrics of all regression models.
2	Regression models were applied to predict the finish time on the actual dataset.	Comparison of the models according to metrics.

TABLE V. EXPERIMENT 1 RESULTS

Model	R^2	MAE	RMSE
Linear Regression	1	0.027	0.039
Lasso Regression	0.998	11.809	16.385
Random Forest	0.979	31.119	53.73
XGBoost	0.968	38.691	67.098
Ridge Regression	0.923	78.769	103.529
LightGBM	0.911	84.732	109.384
Artificial Neural Network	0.848	101.487	144.981
Gradient Boosting	0.846	103.215	145.982
K-Nearest Neighbors	0.763	147.202	181.093
Bayesian Ridge	0.747	152.384	185.672
ElasticNet	0.732	158.903	190.215
Decision Tree	0.721	162.781	193.874
Support Vector Regression	-0.143	329.42	398.004

Its regularization helped reduce overfitting while maintaining accuracy, making it suitable for datasets with many correlated features.

Furthermore, the Random Forest and XGBoost showed strong performance ($R^2 = 0.979$ and 0.968), confirming their effectiveness in handling the relationships of features and being able to handle the mixed data types and missing values in the dataset (e.g., no finish time for the recent 100km). Ridge Regression had a decent performance ($R^2 = 0.923$) but higher error margins compared to Lasso, suggesting that L1 regularization may be more effective for this dataset.

ANN and KNN showed moderate performance, with ANN achieving $R^2 = 0.848$ and KNN $R^2 = 0.763$. These models may require further tuning or feature scaling to improve accuracy. In comparison, LightGBM performs competitively, slightly below Ridge Regression but better than ANN and Gradient Boosting. Gradient Boosting is close to ANN in performance, indicating it can capture non-linear patterns. Bayesian Ridge, ElasticNet, and Decision Tree show moderate performance over the dataset. Finally, SVR performed poorly ($R^2 = -0.143$), showing that it failed to recognize related patterns in the data.

B. Experiment 2

This experiment is to find out whether the regression models perform similarly in the actual finish time dataset based on the training data. According to the results in Table V, models (Linear regression and Lasso regression) were better than the rest of the regression models.

In this experiment, all of the models were applied to a dataset representing the upcoming 100KM trail race, which includes the target race elevation gain and actual finish time. First, the models were used to predict the race time and then compared with the actual time by providing all necessary metrics. Table VII shows the metrics of the regression models.

TABLE VI. REGRESSION MODELS STATISTICS ON THE ACTUAL FINISH TIME DATASET

Rank	Model	MAE (min)	RMSE (min)	R ² Score
1	Support Vector Regression	256.25	294.84	-0.018
2	K-Nearest Neighbors	281.42	392.55	-0.804
3	LightGBM	309.90	423.43	-1.099
4	XGBoost	322.09	437.52	-1.241
5	Decision Tree	326.80	444.80	-1.316
6	Random Forest	352.66	489.76	-1.808
7	Gradient Boosting	361.18	501.22	-1.941
8	Artificial Neural Network	797.59	912.39	-8.747
9	ElasticNet	599.76	962.72	-9.851
10	Linear Regression	633.88	984.37	-10.345
11	Bayesian Ridge	647.55	1009.34	-10.928
12	Lasso Regression	721.76	1145.47	-14.362
13	Ridge Regression	752.30	1209.56	-16.130

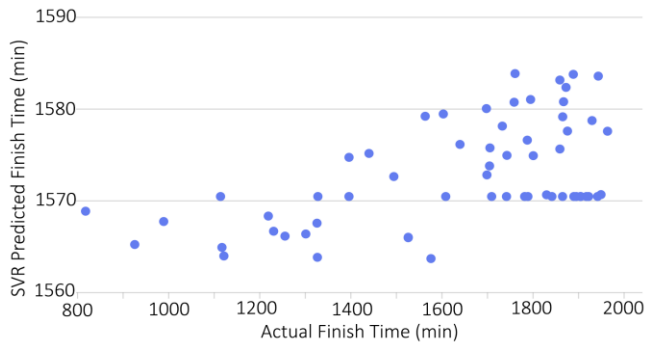


Fig. 1. Plot of SVR Predicted Finish Time vs. Actual Finish Time

Based on the metrics of thirteen regression models using the datasets, the findings reveal significant differences in predictive performance. Experiment 1 shows that Linear regression and Lasso regression perform superior with an almost perfect fit. However, it found that the constructed benchmark dataset was likely to overfit the models. Different results were revealed in Experiment 2; among all models tested, SVR achieved the lowest MAE of 256.25 minutes and the lowest RMSE of 294.84 minutes, with an R^2 score closest to zero (-0.018), indicating better predictive alignment with actual finish times compared to other models.

Fig. 1 depicts how well the SVR prediction compares to the actual finish time, indicating that the SVR has better performance as shown by its low MAE and RMSE. The plot is where each point represents a runner, the x-axis is the actual finish time, and the y-axis is the predicted finish time by the SVR model. The ideal prediction is that the points should be placed close to the diagonal line (where the predicted value is similar to the actual). The SVR model shows a tight clustering around the diagonal, especially for mid-range finish times. Although some deviations for runners with very fast or very slow finish times, however, in overall, the predictions are reasonably aligned.

Close competitors to SVR include K-Nearest Neighbors (KNN), LightGBM, and XGBoost, which also demonstrated relatively low error rates and better generalization compared to other models. These models effectively captured the non-linear relationships and complex interactions present in the target race of 100km ultra-trail running performance data. In contrast, traditional linear models such as Linear Regression, Lasso Regression, and Ridge Regression performed poorly,

with high error metrics and significantly negative R^2 scores. The results suggest that the structured dataset is not well-suited to linear assumptions and may overfit the models. Similarly, the Artificial Neural Network failed to converge effectively, likely due to limited data volume or feature complexity.

Given these results, SVR is recommended as the suitable model for the prediction of the constructed features in the 100km ultra-trail finish times dataset (including a small dataset). It offers balanced performance in both accuracy and robustness, making it a suitable choice for future predictive tasks with similar datasets.

V. CONCLUSION AND FUTURE WORKS

This study explored the experiment of various regression models to predict 100KM ultra-trail finish times using a prepared set of features derived from demographic data, performance indices, race history, timing records, and race-specific elevation gain. The experiments were conducted in two phases: first, evaluating model performance on a training/testing split, and second, applying the trained models to predict actual race outcomes.

The results from Experiment 1 showed that traditional linear models such as Linear Regression, Lasso Regression, and Ridge Regression performed adequately on the training data, with high R^2 scores. However, when applied to the actual race dataset, their performance drops significantly, indicating poor generalization and may overfit the model. In contrast, models like SVR, LightGBM, and XGBoost demonstrated better predictive accuracy on the actual dataset, with SVR achieving the lowest MAE and RMSE. These findings suggest that non-linear models are more suitable for capturing complex relationships in ultra-trail running performance with small data. Surprisingly, although KNN is considered a benchmark algorithm on many machine learning tasks, it performs second to SVR. While not the best model, KNN has shown its simplicity and adaptability on the actual race dataset.

Future research can focus on several enhancements to improve prediction accuracy and model robustness. First, incorporating additional features such as weather conditions, terrain type, and race-specific features could enrich the dataset. Second, exploring ensemble techniques and hybrid models may yield better generalization across diverse race profiles. Third, expanding the dataset to include more runners and events will support deeper learning models like neural networks and XGBoost. Lastly, integrating real-time data streams and adaptive learning could enable dynamic prediction updates during races, offering practical applications for athletes and race organizers during event management.

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